

Exercise 1: Decision Tree

You want to guess whether your friends will join to play beach volley today given the past experiences. You think that 4 elements are important to decide that: the Outlook (O), the Temperature (T), the Humidity (H), and the Wind (W).

You have $O \in \{\text{Sun (S), Overcast (O), Rain (R)}\}$, $T \in \{\text{Warm (W), Mild (M), Cold (C)}\}$, $H \in \{\text{High (H), Normal (N), Low (L)}\}$, and $W \in \{\text{Strong (S), None (W)}\}$.

Here are a table summarizing the previous days:

O	T	H	W	Play?
S	W	H	N	No
S	W	H	S	No
O	W	H	N	Yes
R	M	H	N	Yes
R	C	L	N	Yes
R	C	L	S	No

50]

27, 00]

(s) · gini(s)

$$+ \frac{4}{10} \cdot \frac{2 \cdot 2}{4} + \frac{2 \cdot 2}{10} \cdot \frac{0}{2}$$

2 classes

1 class

1 data

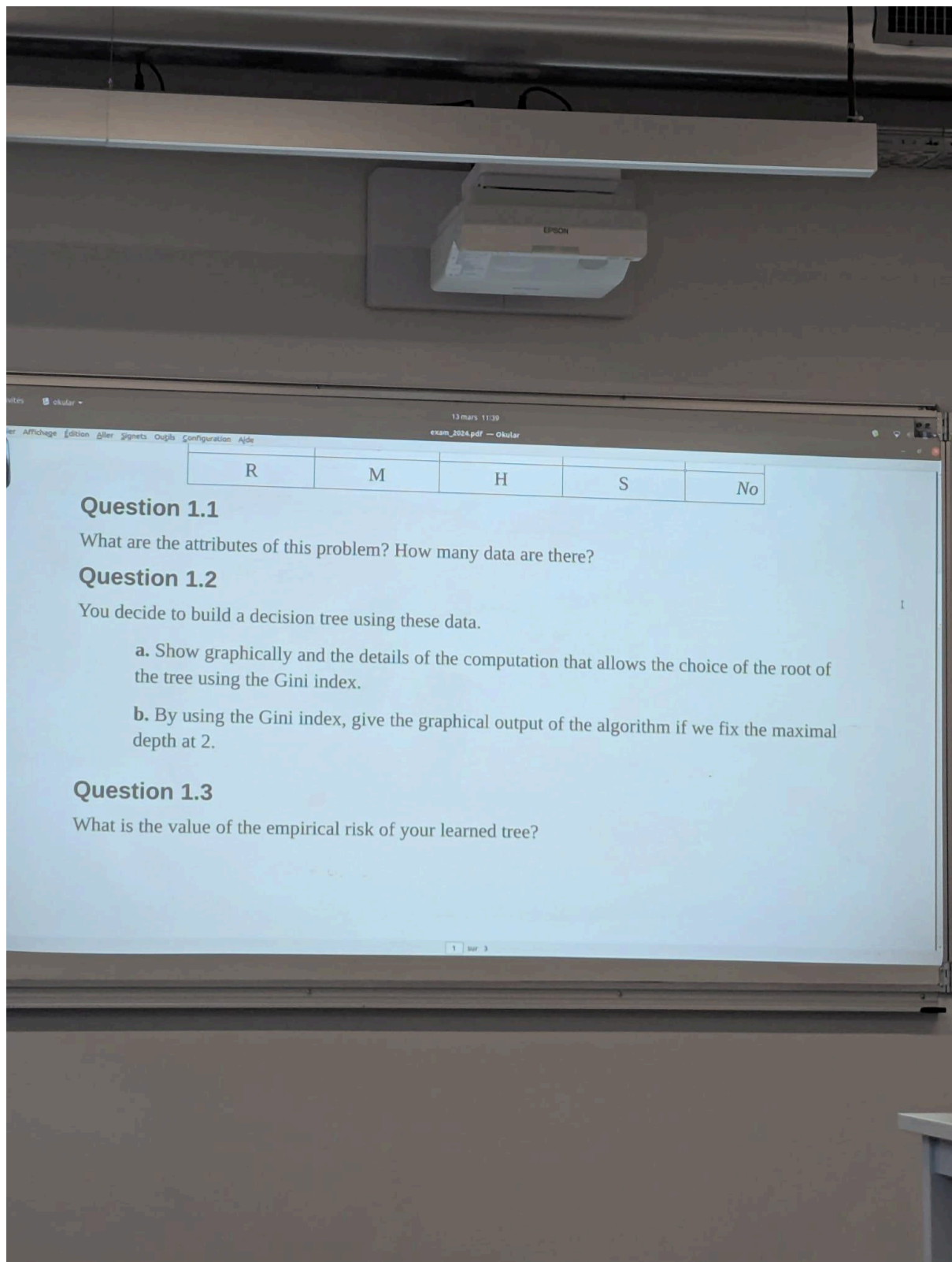
O	T	H	W	Play?
S	W	H	N	No
S	W	H	S	No
O	W	H	N	Yes
R	M	H	N	Yes
R	C	L	N	Yes
R	C	L	S	No
O	C	L	S	Yes
S	M	H	N	No
S	C	L	N	Yes
R	M	H	S	No

Question 1.1

What are the attributes of this problem? How many data are there?

Question 1.2

You decide to build a decision tree using these data.



R	M	H	S	No
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Question 1.1

What are the attributes of this problem? How many data are there?

Question 1.2

You decide to build a decision tree using these data.

- Show graphically and the details of the computation that allows the choice of the root of the tree using the Gini index.
- By using the Gini index, give the graphical output of the algorithm if we fix the maximal depth at 2.

Question 1.3

What is the value of the empirical risk of your learned tree?

Exercise 2: Perceptron

Question 2.1

Recall what a perceptron is in max 2 sentences. What is the goal of a learning algorithm for this model?

Question 2.2

We have these data: $x^1=(-1, -1)$, $y^1=-1$; $x^2=(-1, 1)$, $y^2=+1$; $x^3=(2, 2)$, $y^3=+1$

Run the **dual version** of the perceptron learning algorithm on these data. Do not forget to make the data complete.

To present your computation you should use a table like this one:

	init	First loop			Second loop			...
		x^1	x^2	x^3	x^1	x^2	x^3	
α_1								

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α_1								
α_2								
α_3								

And then detail the computation for each cell below the table.

Question 2.3

Draw the data and the obtained linear separator on a plot.

Question 2.4

Without doing any computation, explain in two sentences max (and an eventual plot) what would happen if we add the data $x^4=(-1, 2)$, $y^4=-1$ and run the learning algorithm again.

happen if we add the data $X^* = (-1, 2)$, $y^* = -1$ and run the learning algorithm again.

Exercise 3: HMM 1

A test based on multiple choice questions consists in selecting the best answers out of the choices from a list to a series of questions. The first objective of this exercise is to model by a HMM the underlying model generating random answers to a set of multiple choice questions.

Let us assume the questions can have either 2, 3, or 4 choices. Thus, the possible answers are $\{a, b\}$, $\{a, b, c\}$ and $\{a, b, c, d\}$, respectively for questions made of 2, 3 and 4 choices.

In this context, a sequence of random answers (e.g. acbda), is a string composed of the successive

Suppose that we have a three states (numbered 1 to 3) HMM λ where the observations can be either a or b such that :

$$A = \begin{pmatrix} 0.4 & 0.3 & 0.3 \\ 0.1 & 0.5 & 0.4 \\ 0.1 & 0.3 & 0.6 \end{pmatrix} \quad B = \begin{pmatrix} 0.0 & 1.0 \\ 1.0 & 0.0 \\ 0.5 & 0.5 \end{pmatrix} \quad \pi = \begin{pmatrix} 0.5 \\ 0.3 \\ 0.2 \end{pmatrix}$$

This means that, for instance, $a_{11}=0.4$ and $b_1(a)=0$.

Question 4.1

Check that $P(ba|\lambda)=0.285$. You should present the computation in the form of a table, as presented in the course.

Question 4.2

What is the most likely sequence of hidden states for the observation ba ? Again, your computation should be presented in the form of a table.

Question 4.3

